

Protein–Ligand Docking Analysis

Description

This project involved performing a basic protein–ligand docking analysis to explore potential interactions between hemoglobin and carbon monoxide (CO).

Method

- Used a molecular docking web server
- Uploaded the hemoglobin protein structure (PDB format)
- Used carbon monoxide as the ligand
- Predicted binding pockets and interaction sites
- Evaluated docking scores and interaction residues

Results

The docking analysis identified several potential binding pockets within the hemoglobin structure.

The best predicted interaction showed a docking score of -2.0 , suggesting a possible binding interaction between carbon monoxide and the protein.

Multiple amino acid residues surrounding the binding region were also identified.

Docking Visualization

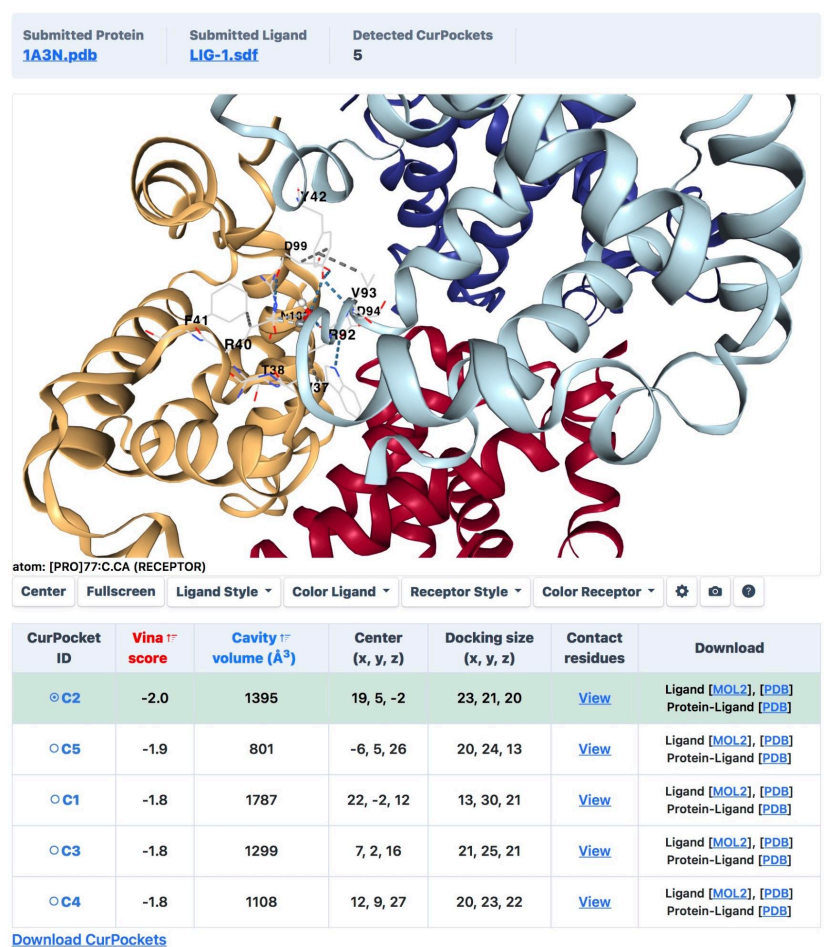


Figure 1: Predicted interaction between hemoglobin and carbon monoxide showing the ligand positioned within a binding pocket of the protein structure.

Interpretation

This analysis demonstrates how molecular docking can be used to predict protein–ligand interactions computationally.

The results provide insight into how small molecules such as carbon monoxide may interact with hemoglobin at specific binding regions.

Although simplified, this workflow introduces an important concept widely used in structural bioinformatics and drug discovery.

Although the docking score was relatively low, this analysis still provided useful insight into how protein–ligand interactions can be explored computationally.

Skills Demonstrated

- Protein–ligand docking
- Structural bioinformatics
- Binding pocket analysis
- Interpretation of docking scores
- Working with PDB structures